Localized Matched-Subspace Detection for Functional Ultrasound and Ultrafast Doppler Imaging

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Abstract

Functional ultrasound is increasingly used in intraoperative, neonatal, and mobile brain imaging, but residual tissue motion, phase instability, and aliased flow can still generate artifactual high scores after conventional clutter suppression. We study localized matched-subspace detectors applied downstream of clutter filtering in beamformed slow-time fUS data, from a fixed band-limited statistic to locally whitened variants. On a held-out SIMUS structural benchmark, changing only the final detector on the same clutter-filtered residual sharply reduces nuisance leakage: in the intra-operative setting, nuisance false-positive rate falls from 0.998 for Kasai to 0.004 for the fixed matched-subspace statistic at matched 50% target-flow recall. In a separate held-out end-to-end benchmark, the selected fixed chains yield zero observed nuisance detections on all four held-out seed-setting pairs. On an open real-IQ rat-brain dataset, the fully whitened variant shows consistent structural improvement across all 10 audited blocks on a conservative ULM vessel-core versus shell evaluation, improving AUC from 0.504 to 0.530. These results show that the downstream detector is a consequential design choice in post-beamformed fUS. Because the method is a drop-in replacement for the final PD/Kasai-style readout on an existing clutter-filtered pipeline, with reported steady-state replay below the acquisition-time budget under the manuscript replay conditions, it offers a practical collaboration path for labs with raw complex IQ or beamformed slow-time data.

1 Introduction

Functional ultrasound (fUS) is now used for vascular and functional mapping in settings ranging from preclinical neuroscience to awake surgery and cranial-window human imaging. In these settings, the practical problem is often not average sensitivity alone, but that even a few artifactual false detections can persist in the Doppler map and complicate interpretation. Tissue motion, phase instability, and aliased high-velocity flow therefore matter most when they survive ordinary clutter suppression and populate the extreme right tail of the detector output.

In intra-operative settings, a false-positive activation region can mislead surgical planning. In resting-state connectivity studies, a single artifactual hot spot that survives across blocks can create a spurious seed correlation. In neonatal bedside monitoring, where acquisition windows are short and motion is unavoidable, artifact leakage in the Doppler map can be hard to distinguish from genuine hemodynamic variation without independent confirmation that is rarely available. These are not average-sensitivity problems; they are rare-event problems driven by the right tail of the score distribution.

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This problem matters now because fUS is no longer only a preclinical proof-of-concept modality. It is already used in intra-operative, neonatal, cranial-window, and emerging mobile settings [1, 8, 9, 21–23], where rare nuisance-driven false positives can matter more than average image quality.

On the held-out SIMUS-Struct benchmark, changing only the downstream detection statistic sharply reduces nuisance leakage on the same clutter-filtered residual. In the intra-operative setting, nuisance FPR falls from 0.998 for Kasai to 0.004 for the fixed matched-subspace statistic at matched 50% target-flow recall. In the separate best-chain benchmark, the selected fixed detector chains yield zero observed nuisance detections on all four held-out seed-setting pairs.

After ordinary clutter suppression, the method asks a local question: within a small neighborhood, is the remaining slow-time signal concentrated in a prespecified flow band, or is it more consistent with structured clutter residual? We study three downstream detection statistics on that fixed residual stream: a fixed matched-subspace statistic, an adaptive variant that selectively invokes local slow-time whitening, and a fully whitened variant that applies whitening throughout. Among these, the fixed statistic is the default on the held-out structural benchmark, while stronger whitening helps in narrower clutter-heavy or structure-concentration settings. Figure 8 later distills that practical choice into a label-free deployment rule. Because the method operates downstream of an existing clutter filter, adopting it does not require a change to the acquisition protocol or to the upstream clutter-filtering pipeline. In practice, a researcher would replace the power-Doppler-style scoring step in an existing pipeline with the matched-subspace statistic on the same clutter-filtered residual and obtain a Doppler map with fewer artifactual false detections. This is a drop-in replacement for the final scoring step of an SVD+PD-style pipeline; acquisition and clutter filtering stay unchanged. In the reported configurations, steady-state replay remains below the acquisition-time budget, so the method is not only a drop-in detector replacement but also computationally practical.

Figure 1 gives a qualitative real-IQ rat-brain preview using the fully whitened variant, which is strongest on this ULM structural endpoint. The default detector choice for general structural use is determined separately by the held-out SIMUS benchmark below.

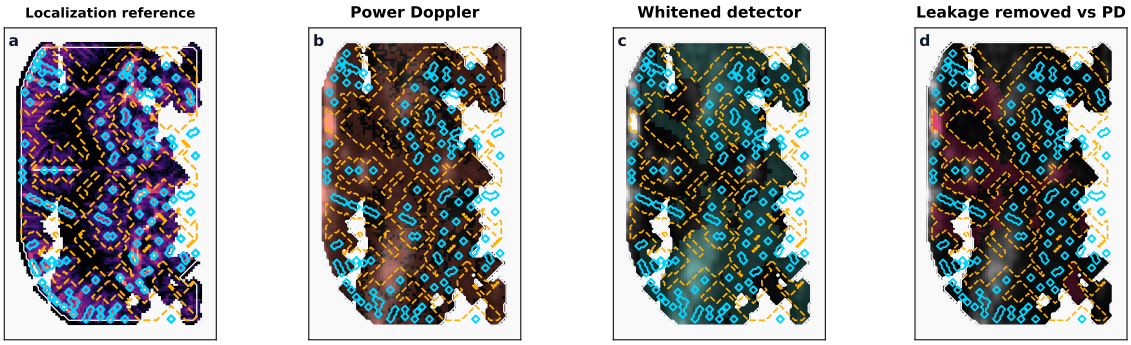

Figure 1: *Illustrative real-IQ structural evaluation preview.* Same audited 64-frame residual window shown with a localization-derived vascular reference, baseline power Doppler, and the fully whitened detector. Panels b–c are thresholded at matched 70% vessel-core recall. Full quantitative same-residual results are reported in Section 5.4; the held-out SIMUS benchmark sets the default detector choice.

We isolate the detector stage after clutter suppression. All detector comparisons use the same clutter-filtered residual so detector effects can be separated from clutter-filter effects.

We evaluate the detector on held-out structural benchmarks, clutter-stress simulations, a structurally labeled phantom, and open real-IQ audits.

Labs with raw complex IQ or beamformed slow-time data, together with task timing, can

evaluate this detector as a drop-in replacement for the final PD/Kasai-style readout within an existing clutter-filtered pipeline. This work provides the detector implementation, same-residual evaluation protocol, and code needed for that comparison.

2 Background and Problem Formulation

Problem setup. For each spatial location (or tile-aggregated snapshot), let $x \in \mathbb{C}^{L_t}$ denote a slow-time vector. We consider

$$H_0 : \; x = c + n, \qquad H_1 : \; x = s + c + n,$$

where s is flow-consistent signal, c is structured clutter/motion, and n is residual noise. In practice, H_0 is spatially nonstationary and heavy-tailed, motivating local covariance adaptation and tail-focused evaluation.

Core notation and bands. We use band operators (P_o, P_f, P_g, P_a) for clutter/DC, flow, guard, and alias regions, respectively. Physically, P_f is the Doppler band where the targeted microvascular flow is expected, P_g is a transition guard band, and P_a captures aliased high-velocity flow or vibration. For short ensembles, these are exact orthogonal DFT-bin projectors; for longer apertures they are leakage-controlled analysis operators and are treated as band-energy statistics with explicit sanity sentinels (Supplementary Material).

2.1 Related context

Spatiotemporal SVD, RPCA, HOSVD, and local adaptive clutter filtering are established in ultrafast Doppler and fUS, and learned clutter filters are relevant prior work on the residual-formation side of the pipeline [11, 12, 31, 32, 45]. These methods improve the residual itself; none explicitly evaluates the downstream detection statistic as a separable design choice on the same residual. The question here is narrower: once a clutter-filtered slow-time cube has been fixed, can the downstream detection statistic reduce artifact leakage more effectively than PD- or Kasai-style readouts? That separates this paper from Yu-, Baranger-, and Song-style local clutter filtering, which adapts the clutter filter itself before a conventional Doppler score is applied [14, 15, 17]. ULM motion correction, twinkling-phantom structure audits, and hemodynamic fUS readouts therefore enter here as evaluation context and downstream audits rather than as the primary algorithmic contribution.

2.2 Baseline scores and detector definitions

Beamformed slow-time data. Let $X[t, y, x] \in \mathbb{C}$ denote a beamformed slow-time IQ movie over $t = 0, \ldots, T-1$ and spatial indices (y, x). We first apply a baseline clutter-suppression operator (e.g. motion-compensated SVD), producing a residual cube $X_{\text{base}}[t, y, x]$ used for both baseline scoring and as the input to the detector family.

Power Doppler proxy and baseline scores. From the baseline-filtered cube we define the baseline ROC score map

$$S_{\text{base}}(y, x) := \frac{1}{T} \sum_{t=0}^{T-1} |X_{\text{base}}(t, y, x)|^2 \,. \tag{1}$$

This is ordinary power Doppler on X_{base}. For short-ensemble settings we additionally report auxiliary conventional comparators such as log-power Doppler and Kasai lag-1 magnitude [16] so that baseline-score idiosyncrasies do not dominate the comparison.
When the same power statistic is used only as a display or proxy map for deterministic preprocessing, we write

$$\text{PD}^{\uparrow}(y, x) \equiv S_{\text{base}}(y, x),$$

to make it explicit that the quantity is being used for visualization or mask construction rather than as the ROC score itself. S_{base} denotes the baseline ROC score map on X_{base}, S_{det} the matched-subspace detector score map, and $\text{PD}^{\uparrow}$ a display or proxy map used only for visualization or deterministic preprocessing rather than for ROC scoring. Unless explicitly stated, the detector family consists of three variants built from the same band-limited statistic family: a fixed matched-subspace statistic without whitening, a prespecified adaptive variant that promotes selected pixels onto a whitened branch when guard-band evidence indicates clutter, and a fully whitened covariance-adaptive variant.

3 Method

What changes relative to SVD + PD. Relative to a conventional ultrafast Doppler pipeline, the proposed detector family changes the final detection statistic rather than replacing upstream clutter suppression. After forming a clutter-filtered slow-time residual, we score its energy inside a prespecified flow subspace rather than using only aggregate Doppler power. The simplest fixed statistic uses that score without local whitening, a prespecified adaptive variant promotes clutter-heavy pixels onto a whitened branch only when guard-band evidence warrants it, and a fully whitened variant is retained for settings where stronger local adaptation is beneficial. The goal is not simply to increase vessel conspicuity, but to reduce nuisance-driven high scores in operating regions where false positives matter.

Plain-language interpretation. The fixed statistic is a local band-limited flow detector applied after ordinary clutter filtering. The adaptive branch adds tile-local whitening only where guard-band energy indicates localized clutter.

Pipeline overview.

1. Baseline clutter suppression (e.g. MC–SVD) produces a residual cube X_{base} and baseline score $S_{\text{base}}(y, x)$.
2. For each tile Ω_i, form the local slow-time support used by the chosen detection statistic; when whitening is active, apply the fixed setting-specific local covariance transform $y_{i,k} = R_i^{-1/2} x_{i,k}$.
3. Compute the self-normalized matched-subspace score on the chosen tile representation and aggregate over overlapping tiles to obtain the detector output map. The fixed and fully whitened forms are given explicitly in Eqs. (2) and (3); setting $R_i = I$ recovers the simpler band-limited fixed statistic.
4. If the adaptive variant is used, retain the fixed statistic by default and promote pixels onto the whitened branch only where a prespecified guard-band clutter feature exceeds a fixed threshold.

In practical terms, the method asks a local question after ordinary clutter suppression: within a small spatial neighborhood, does the remaining slow-time activity look more like microvascular flow

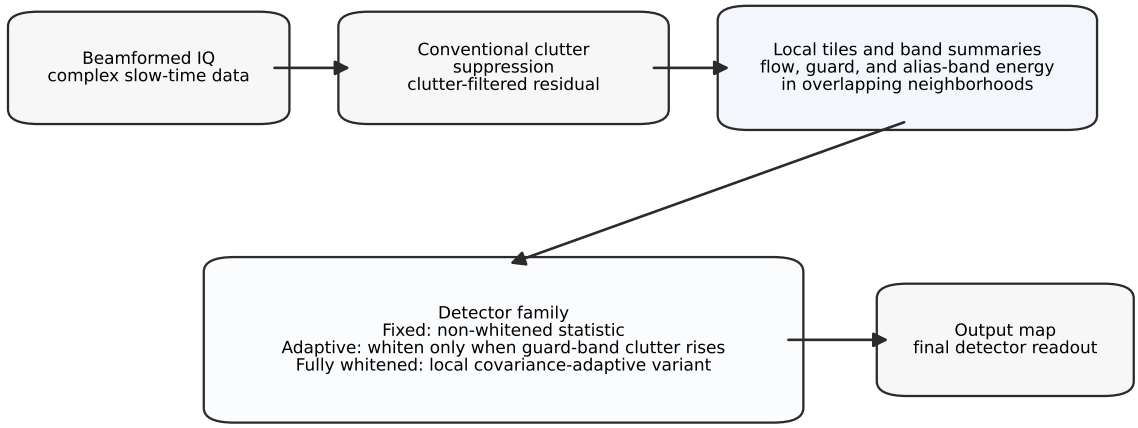

Figure 2: Method overview. Beamformed IQ first undergoes ordinary clutter filtering, after which the detector operates tilewise in slow time. The fixed detector uses the band-limited matched-subspace score everywhere. The adaptive detector uses that same score by default and switches to a whitened branch only in tiles with strong guard-band evidence of localized clutter. The fully whitened detector applies whitening everywhere.

or more like locally structured nuisance? Detector-core ablations on labeled brain simulations and the Gammex phantom are reported in the main Results.

The derivations below formalize the three detector modes introduced above.

Three detector variants. On tile i, let $x_{i,k}$ denote the unwhitened slow-time snapshots and $y_{i,k} = R_i^{-1/2} x_{i,k}$ the whitened snapshots when whitening is active. The fixed detector is

$$S_0(i) := \frac{1}{K} \sum_{k=1}^{K} \frac{x_{i,k}^{\mathrm{H}} P_f \, x_{i,k}}{x_{i,k}^{\mathrm{H}} (I - P_f) \, x_{i,k} + \rho \, x_{i,k}^{\mathrm{H}} x_{i,k} + \epsilon}, \tag{2}$$

which is the same band-limited matched-subspace ratio with $R_i = I$. The fixed detector rewards energy in the slow-time band where microvascular flow is expected and penalizes residual energy elsewhere; whitening is added only when the local residual looks clutter-dominated.

$$S_w(i) := \frac{1}{K} \sum_{k=1}^{K} \frac{y_{i,k}^{\mathrm{H}} P_f \, y_{i,k}}{y_{i,k}^{\mathrm{H}} (I - P_f) \, y_{i,k} + \rho \, y_{i,k}^{\mathrm{H}} y_{i,k} + \epsilon}. \tag{3}$$

In plain language, the numerator measures energy in the expected flow band, while the denominator measures energy everywhere else plus a small stabilizing ridge; the ratio is high only when flow-band energy dominates the rest of the slow-time spectrum. The detector family has three concrete

variants. The fixed detector uses $S_0(i)$ everywhere. The prespecified adaptive detector uses

$$S(i) = \begin{cases} S_w(i), & r_g^{\text{base}}(i) \geq \tau_g, \\ S_0(i), & r_g^{\text{base}}(i) < \tau_g, \end{cases} \tag{4}$$

where $r_g^{\text{base}}(i)$ is a baseline guard-band energy fraction and τ_g is fixed once and then held fixed across all reported experiments. The fully whitened detector uses $S_w(i)$ everywhere. In words, the adaptive detector keeps the simpler score in clean tiles and only turns on full whitening where the residual itself shows guard-band evidence of localized clutter.

Tile support and switching rule. All reported detectors operate on overlapping local tiles. In the main quantitative settings, the spatial support is fixed at 8×8 pixels, with slow-time aperture $L_t = 8$ for kWave-Stress, $L_t = 16$ for Gammex-Struct, and $L_t = 64$ for SIMUS-Struct and ULM-CoreShell. The adaptive rule evaluates the baseline guard-band fraction $r_g^{\text{base}}(i)$ on each tile and switches from $S_0(i)$ to $S_w(i)$ only when $r_g^{\text{base}}(i) \geq \tau_g$; otherwise the fixed score is retained.
Proxy construction, band discretization, and conditional-execution checks are deterministic. Invalid configurations are excluded from reporting, and decision-rule variants such as conditional execution are reported separately from the default detector pipeline.

4 Experimental Design

4.1 Prespecified operating protocol and operating-point reporting

All processing settings (tile geometry, aperture, band definitions, and adaptive threshold) are fixed for each setting before held-out results are reported. Detection is evaluated by thresholding the high-score tail (right tail) at fixed FPR targets:

$$\mathbb{P}\{S \geq \tau_\alpha \mid H_0\} = \alpha, \qquad \text{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}.$$

For each experiment, we report the supported floor $fpr_{\min} = 1/n_{\text{neg}}$ and bootstrap uncertainty at the natural unit of variation (windows or frames, depending on setting). These choices enforce a strict low-FPR operating discipline and avoid over-interpreting unsupported tail points. We emphasize $\alpha = 10^{-3}$–10^{-4} because those operating points expose nuisance-tail failure modes cleanly, not because routine fUS use must literally operate at those exact values. The same map-trustworthiness problem is already visible at a conventional matched background FPR of 10^{-2} in the worked Gammex example of Figure 6, where the baseline thresholded map is dominated by off-lumen detections while the matched-subspace score recovers lumen-consistent structure.

Threshold calibration and deployment. Unless otherwise stated, τ_α is calibrated per window (or per cine sequence) on that unit's negative set to report TPR(α) at the intended operating point; in labeled settings, the negative set is available for *exact* calibration, while in deployment it would be estimated via a negative-only training procedure or from a fixed calibration set. The Results section therefore includes two non-retrospective transfer checks: a fixed-calibration audit on SIMUS-Struct and a complementary block-split fixed-calibration audit on the ULM structural evaluation.

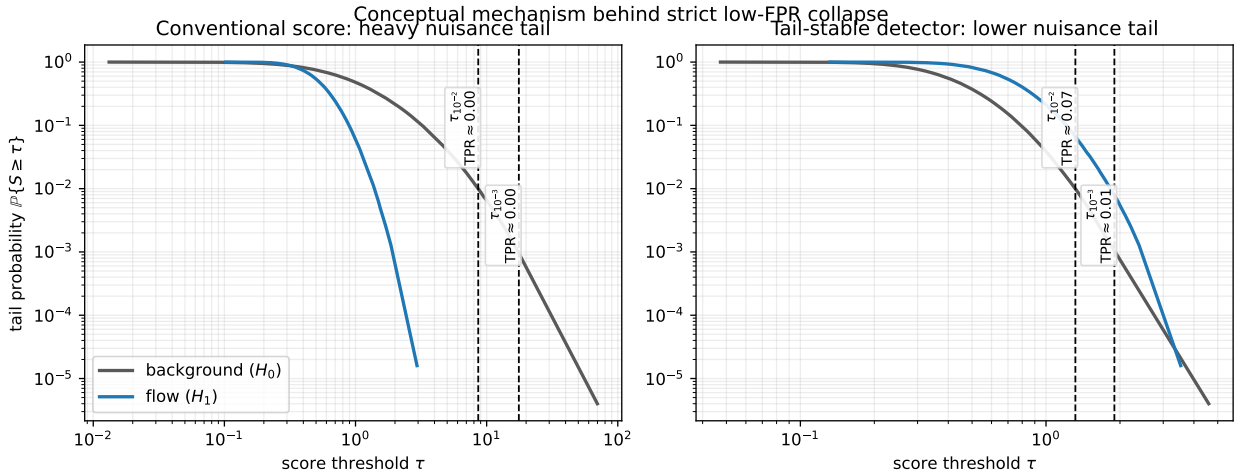

Figure 3: Conceptual strict-tail mechanism. Left: a conventional score can still show moderate-FPR separation while failing at a stricter operating point because the background right tail pushes the background-calibrated threshold $\tau_{10^{-3}}$ above essentially all flow pixels, yielding TPR ≈ 0 by construction. Right: when the nuisance tail is reduced, the same strict background allowance leaves some flow pixels above threshold. The later labeled-brain and Gammex tail figures are dataset-specific examples of this same mechanism.

How the prespecified defaults were chosen. Band edges are tied to acquisition physics, while tile geometry and L_t are chosen once per setting to preserve spatial adaptation with stable detector estimates and then held fixed. The adaptive threshold τ_g is a single frozen guard-energy cutoff chosen to keep the fixed score in cleaner settings and enable whitening only when guard-band clutter evidence is clearly non-negligible.
Additional checks for acquisition metadata and compute gating are deterministic and are applied before reporting.

4.2 Regimes and scope

The evidence is stratified by setting so that quantitative detection results remain clearly separated from robustness or transfer audits. We use the short labels kWave-Stress, Gammex-Struct, SIMUS-Struct-Mobile, SIMUS-Struct-Intraop, ULM-CoreShell, ULM-Motion, and Shin-ProxyMotion throughout; SIMUS-Struct denotes the paired mobile and intra-operative structural benchmark. Downstream detection-statistic design matters after a fixed clutter filter; the fixed band-limited statistic is the default structural choice on the held-out SIMUS benchmark; adaptive and fully whitened variants help in narrower clutter-heavy or structure-concentration settings; and fixed-threshold transfer is encouraging in selected settings but not solved.

Each setting uses one frozen operating profile for baseline, detector support, and Doppler-band geometry.

Equation (1) defines the baseline ROC score, Equation (2) the fixed detector, Equation (4) the adaptive detector, and Equation (3) the fully whitened detector. Setting-specific whitening estimators, loading values, and training-support policies are fixed before reporting.

5 Results

Results roadmap. We begin with the held-out SIMUS benchmark and proceed through mechanism checks and real-IQ audits.

5.1 SIMUS-Struct benchmark

SIMUS-Struct uses the open-source SIMUS/PyMUST moving-scatterer forward model and is calibrated to reported fUS operating conditions rather than to a generic parameter sweep [8,9,35,46–48]. We report two prespecified structural settings, SIMUS-Struct-Mobile and SIMUS-Struct-Intraop, with frozen positive, background, and nuisance masks and held-out evaluation on seeds 127–128 after development on seeds 125–126. The positive class is unaliased parenchymal Pf-band flow; the negatives are ordinary background and explicit nuisance structures such as superficial high-velocity or specular components. Throughout this section, $AUC_{\text{main/bg}}$ measures separation from background, $AUC_{\text{main/nuis}}$ measures separation from clinically relevant nuisance, and FPR_{nuis} @ $TPR_{\text{main}} = 0.5$ measures nuisance leakage when half of the target positives are retained.

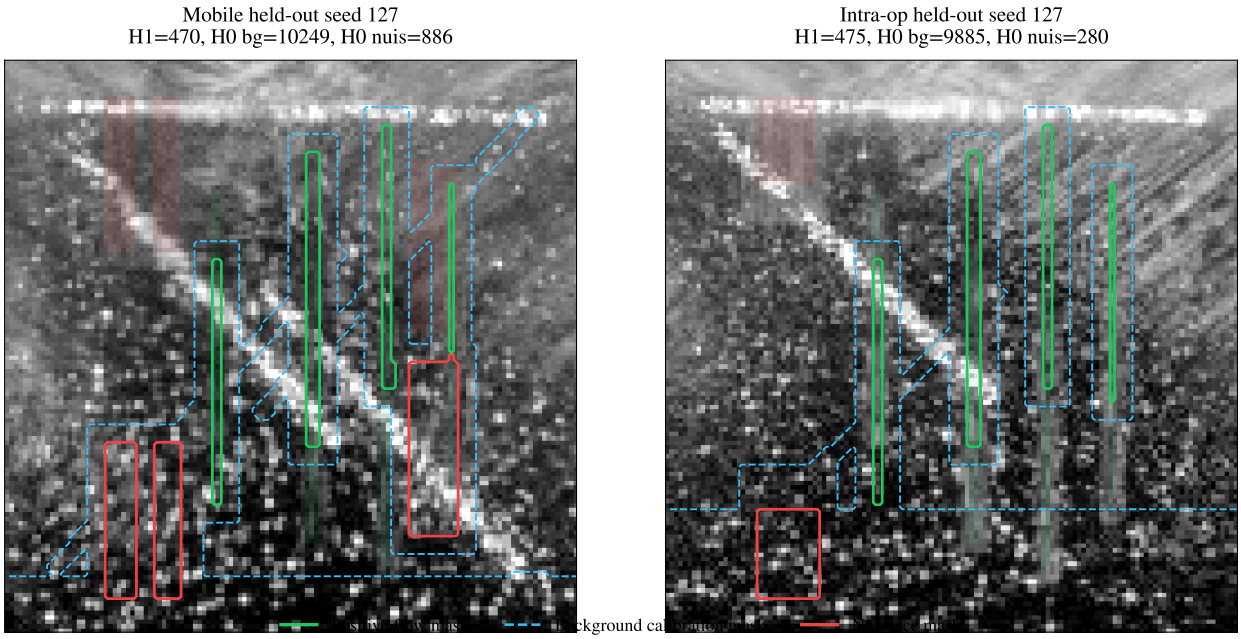

Figure 4: Representative SIMUS-Struct geometry for the SIMUS-Struct-Mobile and SIMUS-Struct-Intraop reference settings. The grayscale underlay is log beamformed power from the simulated IQ cube and is shown only to make the mask geometry visible. Green contours mark the parenchymal-flow positive set H_1, blue dashed contours mark the broad background pool H_0^{bg} used to set the score threshold, and red contours mark the nuisance-negative set H_0^{nuis} used to quantify structured false detections at that same operating point. Table 2 reports the held-out seed results on these frozen mask types across seeds 127–128.

For the functional task, we use the same prespecified settings and a common background-controlled GLM-style readout on simulated task/null runs. We refer to that experiment as the

SIMUS functional audit. Figure 4 shows the corresponding structural mask geometry, and Table 2 gives the held-out reference setting that determines the default downstream score.

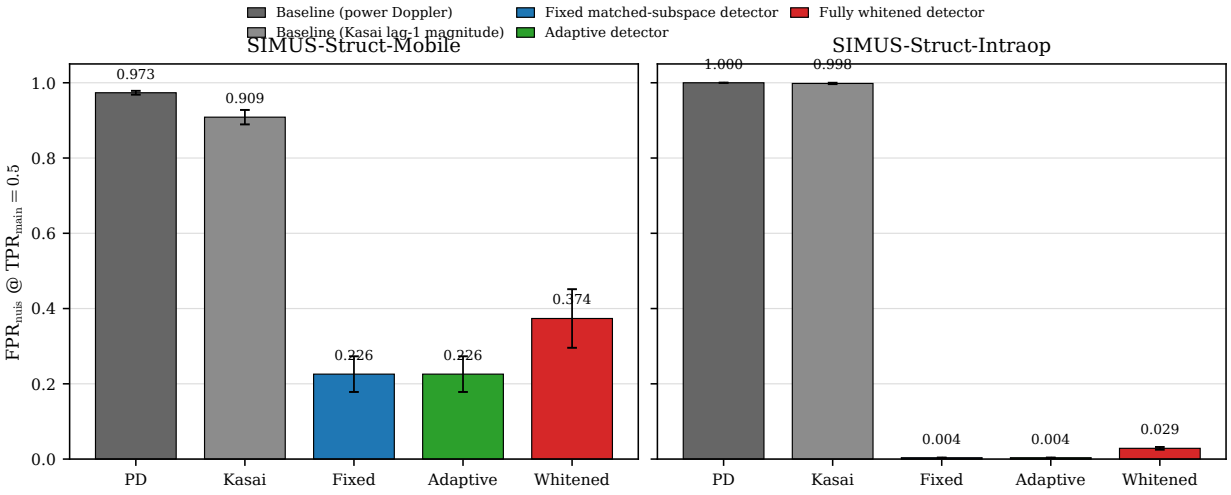

Figure 5: Headline same-residual scoring-rule comparison on the held-out SIMUS reference benchmark using the common MC–SVD residual from Table 3. Bars show nuisance false-positive rate at matched main-response recall 0.5, the clearest operating-point summary of the benchmark. Lower bars are better. On the intra-operative setting, the fixed matched-subspace detector reduces nuisance FPR from 0.998 for Kasai to 0.004 on the same residual stream; under this frozen guard rule, the adaptive variant matches that result. $\mathrm{AUC}_{\mathrm{main/nuis}}$ values are reported in Table 3.

Table 2 is the best-chain benchmark: each row reports the strongest held-out end-to-end pipeline for that setting, so both clutter filter and final detector may differ across rows.

Mobile

Best conventional baseline: HOSVD → Kasai. Fixed detector chain: Adaptive Global SVD → fixed matched-subspace.

Summary	Best conventional baseline			Fixed detector chain		
	$AUC_{main/bg}$	$AUC_{main/nuis}$	FPR_{nuis} @ $TPR_{main} = 0.5$	$AUC_{main/bg}$	$AUC_{main/nuis}$	FPR_{nuis} @ $TPR_{main} = 0.5$
Mean [min,max] over seeds 127–128	0.669 [0.647,0.692]	0.672 [0.624,0.720]	0.286 [0.210,0.361]	0.851 [0.798,0.905]	0.953 [0.923,0.984]	$0.000^{\dagger}$ [0.000,0.000]

Intra-operative parenchymal

Best conventional baseline: HOSVD → Kasai. Fixed detector chain: RPCA → fixed matched-subspace.

Summary	Best conventional baseline			Fixed detector chain		
	$AUC_{main/bg}$	$AUC_{main/nuis}$	FPR_{nuis} @ $TPR_{main} = 0.5$	$AUC_{main/bg}$	$AUC_{main/nuis}$	FPR_{nuis} @ $TPR_{main} = 0.5$
Mean [min,max] over seeds 127–128	0.837 [0.834,0.839]	0.451 [0.431,0.471]	0.621 [0.600,0.643]	0.994 [0.993,0.995]	0.99996 [0.99994,0.99998]	$0.000^{\dagger}$ [0.000,0.000]

†Exactly zero observed nuisance detections; supported floor 1/886 (mobile) and 1/280 (intra-operative parenchymal).

Table 2: Held-out SIMUS-Struct reference settings. For each evaluated structural setting, we report the best conventional baseline from the development-split baseline sweep and the strongest fixed detector chain chosen on development seeds 125–126, then evaluated unchanged on held-out seeds 127–128. Values are mean [min,max] over the two held-out seeds. $AUC_{main/bg}$ measures separation of target flow from ordinary background, $AUC_{main/nuis}$ measures separation of target flow from explicit nuisance structures, and FPR_{nuis} @ $TPR_{main} = 0.5$ asks how many nuisance detections survive when the detector is tuned to retain half of the target positives.

Best-chain benchmark. On the mobile setting, HOSVD → Kasai is the strongest conventional baseline. Adaptive Global SVD → fixed matched-subspace improves $AUC_{main/nuis}$ from 0.672 [0.624, 0.720] to 0.953 [0.923, 0.984] and yields zero observed nuisance detections on both held-out seeds. On the intra-operative parenchymal setting, HOSVD → Kasai is again the strongest conventional baseline, while RPCA → fixed matched-subspace improves $AUC_{main/nuis}$ from 0.451 [0.431, 0.471] to 0.99996 [0.99994, 0.99998] and again yields zero observed nuisance detections on both held-out seeds. Figure 4 shows the held-out mask geometry used for these comparisons, and the 0.000 nuisance rows in Table 2 are literal zero observed nuisance detections rather than rounded near-zero rates. The detector-only same-residual comparison is reported separately in Table 3 and Figure 5. This is the benchmark result that determines the default detector configuration.

Same-residual scoring-rule ablation. To isolate that detection-statistic claim directly on the main benchmark, Table 3 fixes one held-out MC–SVD residual stream and swaps only the downstream detection statistic. This removes clutter-filter choice from the comparison. On that fixed residual, PD and Kasai still rank nuisance highly, the fixed matched-subspace statistic is the strongest same-residual detector, and the fully whitened statistic is worse on both held-out settings. The adaptive statistic is numerically identical to the fixed statistic here because its frozen guard-promotion rule does not activate on these cleaner MC–SVD residuals. On this benchmark, the adaptive extension therefore preserves the fixed default when the clutter evidence is weak instead of forcing whitening where it does not help.
Table 3 is the same-residual score-swap comparison: every row uses the identical MC–SVD residual cube, and only the final scoring rule changes.

Method	Mobile			Intra-operative parenchymal		
	$AUC_{main/bg}$	$AUC_{main/nuis}$	FPR_{nuis} @ $TPR_{main} = 0.5$	$AUC_{main/bg}$	$AUC_{main/nuis}$	FPR_{nuis} @ $TPR_{main} = 0.5$
Baseline (power Doppler)	0.568 [0.562,0.575]	0.131 [0.127,0.134]	0.973 [0.968,0.979]	0.912 [0.909,0.915]	0.022 [0.016,0.028]	1.000 [1.000,1.000]
Baseline (Kasai lag-1 magnitude)	0.500 [0.489,0.512]	0.171 [0.154,0.188]	0.909 [0.889,0.928]	0.792 [0.788,0.795]	0.019 [0.017,0.022]	0.998 [0.996,1.000]
Fixed matched-subspace detector	0.623 [0.587,0.659]	0.713 [0.677,0.748]	0.226 [0.178,0.273]	0.844 [0.831,0.857]	0.982 [0.981,0.983]	0.004 [0.004,0.004]
Adaptive detector	0.623 [0.587,0.659]	0.713 [0.677,0.748]	0.226 [0.178,0.273]	0.844 [0.831,0.857]	0.982 [0.981,0.983]	0.004 [0.004,0.004]
Fully whitened detector	0.569 [0.553,0.585]	0.600 [0.529,0.672]	0.374 [0.296,0.451]	0.819 [0.819,0.820]	0.932 [0.927,0.936]	0.029 [0.025,0.032]

Table 3: Same-residual detector-family ablation on SIMUS-Struct using a common MC–SVD residual. Rows differ only in the downstream scoring rule: power Doppler, Kasai lag-1 magnitude, the fixed flow-band matched-subspace detector without whitening ($R = I$), the adaptive detector that promotes clutter-heavy tiles onto a whitened branch, and the fully whitened detector. Values are means with [min,max] over held-out evaluation seeds 127 and 128 for each setting. Lower nuisance FPR at matched TPR is better.

FPR_{nuis} @ $TPR_{\text{main}} = 0.5$ directly measures artifact leakage at a clinically relevant sensitivity operating point.

A fixed-threshold transfer audit also survives on the same benchmark family: on SIMUS-Struct-Mobile, the fixed statistic preserves nonzero held-out main recall while reducing realized nuisance FPR relative to PD, and on SIMUS-Struct-Intraop it is the only statistic that suppresses nuisance leakage to zero under the transferred threshold.

Clutter-heavy stress setting. Whitening starts to matter once the same benchmark is pushed into clutter-heavier settings. We test two held-out stresses: stronger cardiac-like low-frequency pulsation and shorter slow-time support (64, 48, and 32 frames). The pulsation rows use a fixed 1.30 Hz nuisance term with increasing amplitude, and the short-ensemble rows mimic clinically constrained acquisitions in which shorter windows are used to limit motion corruption or workflow burden [1, 11, 12, 23, 35]. The held-out evaluation is then rerun at full reporting tier for the clutter-filter families that survived the reduced-grid search.

Cardiac-like pulsation

Level / perturbation	Baseline (RPCA→PD)	Fixed statistic	Adaptive statistic	Whitened variant
Ref.	$0.381^{0.416}_{0.345}$	$0.879^{0.898}_{0.860}$	$0.879^{0.898}_{0.860}$	$0.911^{0.925}_{0.894}$
1.0x cardiac amplitude at 1.30 Hz	$0.743^{0.842}_{0.652}$	$0.034^{0.047}_{0.023}$	$0.034^{0.046}_{0.023}$	$0.017^{0.027}_{0.008}$
Mod.	$0.388^{0.423}_{0.354}$	$0.877^{0.895}_{0.857}$	$0.877^{0.895}_{0.857}$	$0.903^{0.920}_{0.886}$
2.0x cardiac amplitude at 1.30 Hz	$0.668^{0.795}_{0.593}$	$0.049^{0.064}_{0.030}$	$0.049^{0.064}_{0.030}$	$0.030^{0.043}_{0.019}$
Hard	$0.399^{0.433}_{0.365}$	$0.875^{0.893}_{0.854}$	$0.875^{0.894}_{0.854}$	$0.905^{0.921}_{0.890}$
3.0x cardiac amplitude at 1.30 Hz	$0.609^{0.770}_{0.537}$	$0.044^{0.058}_{0.029}$	$0.044^{0.059}_{0.029}$	$0.027^{0.041}_{0.017}$

Short ensemble

Level / perturbation	Baseline (RPCA→PD)	Fixed statistic	Adaptive statistic	Whitened variant
Ref.	$0.381^{0.415}_{0.345}$	$0.879^{0.897}_{0.861}$	$0.879^{0.898}_{0.859}$	$0.911^{0.926}_{0.895}$
64 frames (42.7 ms)	$0.743^{0.840}_{0.649}$	$0.034^{0.046}_{0.023}$	$0.034^{0.047}_{0.023}$	$0.017^{0.027}_{0.008}$
Mod.	$0.390^{0.423}_{0.356}$	$0.784^{0.807}_{0.758}$	$0.784^{0.808}_{0.759}$	$0.846^{0.868}_{0.824}$
48 frames (32.0 ms)	$0.754^{0.814}_{0.708}$	$0.137^{0.169}_{0.102}$	$0.137^{0.167}_{0.103}$	$0.058^{0.080}_{0.034}$
Hard	$0.426^{0.462}_{0.392}$	$0.772^{0.796}_{0.746}$	$0.772^{0.797}_{0.746}$	$0.716^{0.745}_{0.686}$
32 frames (21.3 ms)	$0.617^{0.727}_{0.536}$	$0.166^{0.194}_{0.139}$	$0.166^{0.193}_{0.139}$	$0.160^{0.207}_{0.130}$

Table 4: Held-out SIMUS mobile stress setting with RPCA fixed as the clutter filter on every row. Entries report $\mathrm{AUC}_{\mathrm{main/nuis}}$ (top) and $\mathrm{FPR}_{\mathrm{nuis}}$@0.5 (bottom), with point estimates shown as $x^{\mathrm{hi}}_{\mathrm{lo}}$ using 95% nonparametric bootstrap intervals over the held-out masked score samples. RPCA is used because RPCA $\rightarrow$ PD is the strongest conventional baseline among evaluated methods on every reported stress row. This table separates the fixed, adaptive, and fully whitened detector statistics directly instead of reporting only the best family member on each row.

Table 4 shows the punchline directly: the fully whitened statistic is strongest on the cardiac-pulsation rows and on the 64-frame and 48-frame short-ensemble rows, while the fixed statistic remains strongest only on the hardest 32-frame row. The adaptive statistic is numerically identical to the fixed statistic on these rows, so the gain is not coming from freely switching among all three modes. Because this stress setting uses one held-out evaluation seed per axis, the uncertainty notation here is a within-seed bootstrap interval over the held-out masked scores rather than a between-seed interval.

Functional same-cube detection-statistic comparison. A held-out functional audit on the same residual stream is directionally positive for the fully whitened variant across most residual families, but the effect is smaller than in the structural benchmark and is treated as secondary evidence.

Scope and low-FPR operating protocol. Quantitative detection performance is reported on kWave-Stress, Gammex-Struct, and SIMUS-Struct using the strict low-FPR operating protocol of Section 4.1. The ULM and Shin sections add structural, robustness, and transfer audits on open real-IQ data. Conditional execution is a decision-rule variant that uses a flow-proxy mask; leakage ablations and additional diagnostics are reported in Supplementary Material.

5.2 Brain simulation stress tests

The held-out SIMUS-Struct benchmark already determines the default downstream statistic, so the labeled brain simulations are kept brief here. These controlled kWave-Stress audits use labeled brain simulations generated with the k-Wave wave-propagation toolbox [13] and ask a narrower mechanism question on the same MC–SVD residual cube: does the gain come from band selectivity, from whitening alone, or from their combination? The answer is that whitening alone is not sufficient in these stress tests. On fixed residuals, whitened total power collapses at the reported strict-tail operating points, whereas the fixed band-limited matched-subspace score remains strong; detector-swap audits also show that gains do not automatically persist on more aggressive clutter filters and that the fully whitened variant remains stable under modest covariance-training trim changes. Taken together, these labeled stress tests explain why the held-out SIMUS benchmark favors the fixed statistic and why whitening becomes useful only in narrower clutter-heavy settings.

5.3 Gammex phantom: structural ROC and within-ensemble motion ladders

Dataset and acquisition parameters. We use the open-access Doppler Twinkling Artifact database [43], which provides complex beamformer output from phantom acquisitions on a clinical scanner. The *Gammex flow phantom* provides a simple vessel-like geometry that lets us derive independent structure-based flow/background labels from B-mode anatomy alone. Each cine frame

contains a contemporaneous B-mode image and a short-ensemble color-flow (CFM) slow-time block with $N \in \{5, 9, 17\}$ pulses. Leonov *et al.* explicitly report PRF for some acquisition families, including the transverse linear-probe Gammex acquisition, but also note that PRF information is not consistently preserved across the database. We therefore fix PRF once per sequence family, using the source-paper value where it is explicitly documented and otherwise recording a sequence-level override in bundle metadata. Band-discretization sanity checks flag invalid Pf/Pa discretizations explicitly. Decoding/QC details are provided in Supplementary Material (PRF provenance and decoder sanity); reproduction commands and manifests are provided in Supplementary Material.

B-mode-only structural tube masks (independent H_1/H_0). For the Gammex flow phantom sequences we derive evaluation masks from B-mode structure only (no Doppler or matched-subspace outputs). Using acquisition geometry metadata that specifies the CFM region-of-interest (ROI) within the B-mode grid (beam and sample offsets and any beam subsampling), we map the B-mode ROI onto the CFM grid and segment the tube lumen on a median log-envelope reference image. This yields (i) a conservative lumen (flow) mask, eroded to avoid wall pixels and (ii) a background mask that excludes a guard band around the lumen and bright reflectors. Mask provenance and QC metrics are recorded with each experiment. These structural labels support a defensible phantom ROC.

Frozen short-ensemble Doppler profile and baseline fairness scores. For the linear-probe Gammex sequences used here ($N = 17$ shots; PRF values listed in Supplementary Material), we use a prespecified short-ensemble operating profile: baseline SVD band-pass keep range $[2, 17]$, detector aperture $L_t = 16$, tiles 8×8, a Tyler-type robust covariance estimator, and diagonal loading $\lambda = 0.07$. To keep the profile auditable while avoiding sample-support limitations on small ROIs and excessive overlap/mixing on large ROIs, we use a deterministic geometry-only stride policy: choose the largest tile stride ≤ 6 such that the implied tile count is ≥ 500. On the two Gammex views this yields stride 6 (along; large ROI) and stride 4 (across; small ROI). Band definitions are set to $P_f = [150, 450]$ Hz and $P_a = [700, 1200]$ Hz. For baseline context on short ensembles, we also report conventional comparators derived from the baseline-filtered cube (power Doppler, log power Doppler, and Kasai lag-1 power). Reproducibility details are provided in Supplementary Material.

Structural-label operating points (Gammex). The same strict-tail geometry of Figure 3 appears in the short-ensemble Gammex phantom, where residual twinkling structure dominates the upper background tail. We therefore pool pixel scores across cine frames and report low-FPR operating points using right-tail thresholding on pooled background pixels, then evaluate TPR on pooled lumen pixels. In these short-ensemble Doppler acquisitions, the background set includes strong artifact pixels whose right-tail scores dominate conventional baselines; consequently, low-FPR thresholds at $\alpha \leq 10^{-3}$ can lie above essentially all lumen pixels, yielding numerically zero baseline TPR at the reported operating points. To avoid treating pooled pixels as independent samples, we report 95% confidence intervals via a frame-level bootstrap: we resample cine frames as the unit of variation while holding the pooled threshold fixed. Table 5 summarizes a broader same-residual score comparison for the two Gammex views under those same settings: power Doppler and Kasai on the baseline residual, plus three detector-family score variants evaluated on that identical residual cube (whitened power, the fixed matched-subspace detector, and the fully whitened

matched-subspace detector). For compactness we omit log-PD from the main table because it is monotone-equivalent to PD here and numerically identical at the reported operating points; the full pooled summaries remain available in Supplementary Material. The practical relevance is already visible at a conventional operating point. On the representative Figure 6 frame at matched background FPR 10^{-2}, the baseline thresholded map contains 884 suprathreshold pixels and none lie inside the structural lumen mask, whereas the matched-subspace score recovers 586 lumen pixels while holding the same background allowance. Table 5 now reports that relaxed $\alpha = 10^{-2}$ point alongside the stricter tail operating points, so the phantom result can be read both as a practitioner-facing map-quality summary and as a stress test. The same-residual phantom comparison shows that the baseline residual still contains recoverable lumen signal, because changing only the downstream score restores nonzero sensitivity. In this short-ensemble setting, local whitening is the dominant ingredient, and total whitened slow-time power is stronger than the matched-subspace ratio on both views, especially across-view. Gammex therefore supports localized covariance-aware downstream scoring, but not the claim that the matched-subspace ratio is the preferred score in every phantom setting. The view asymmetry still matters: the across view provides the clearest structural-label phantom evidence, whereas the along view remains a harsher artifact-tail stress test under this short-ensemble profile. A small untuned sanity audit in Supplementary Material gives similar values: on the first 20 cine frames, applying the matched-subspace detector directly to raw registered IQ changes TPR@10^{-3} only from 3.48×10^{-4} to 2.61×10^{-4} on the along view and from 0.202 to 0.205 on the across view; perturbing the frozen along-view PRF from 2000 to 3000 Hz leaves TPR@10^{-3} unchanged at 3.48×10^{-4} in that limited audit.

	Along (linear17 @ PRF 2500)				Across (linear17 @ PRF 2500)			
Method / score	TPR@$\alpha = 10^{-2}$	TPR@$\alpha = 10^{-4}$	TPR@$\alpha = 3\times10^{-4}$	TPR@$\alpha = 10^{-3}$	TPR@$\alpha = 10^{-2}$	TPR@$\alpha = 10^{-4}$	TPR@$\alpha = 3\times10^{-4}$	TPR@$\alpha = 10^{-3}$
Baseline (power Doppler)	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]
Baseline (Kasai lag-1 power)	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]
Detector ablation (whitened power)	**0.4116 [0.4078,0.4155]**	**0.0012 [0.0011,0.0014]**	**0.0119 [0.0114,0.0124]**	**0.0532 [0.0521,0.0543]**	**0.7234 [0.7206,0.7262]**	**0.4820 [0.4780,0.4863]**	**0.5155 [0.5115,0.5191]**	**0.5813 [0.5783,0.5843]**
Detector ablation (unwhitened ratio)	0.00031 [0.00022,0.00041]	0.00000 [0.00000,0.00001]	0.00000 [0.00000,0.00001]	0.00001 [0.00000,0.00002]	0.1095 [0.1040,0.1148]	0.0574 [0.0533,0.0611]	0.0602 [0.0562,0.0643]	0.0653 [0.0610,0.0695]
Fully whitened matched-subspace detector	0.0942 [0.0926,0.0958]	0.00000 [0.00000,0.00000]	0.00003 [0.00000,0.00007]	0.00031 [0.00020,0.00043]	0.6891 [0.6846,0.6939]	0.0491 [0.0469,0.0513]	0.0998 [0.0970,0.1027]	0.2237 [0.2197,0.2276]

Table 5: Gammex flow phantom structural-label operating points using B-mode-derived structural masks. All scores are right-tail (higher indicates more flow evidence). Every row is evaluated on the identical baseline SVD band-pass residual cube; only the final scoring rule changes. Rows compare power Doppler, Kasai, whitened power, an unwhitened matched-subspace ratio, and the fully whitened matched-subspace detector on that same residual. The table includes $\alpha = 10^{-2}$ as a practitioner-facing relaxed operating point together with the stricter $\alpha \leq 10^{-3}$ tail regime used for stress testing. Thresholds are chosen per method on pooled background pixels to match each target FPR and evaluated on pooled lumen pixels. Brackets show 95% frame-bootstrap CIs ($n = 2000$ resamples over cine frames) at the fixed pooled threshold. Pooled background counts are $n_{\mathrm{bg}} = 3.85 \times 10^6$ for the along view ($fpr_{\min} \approx 2.6 \times 10^{-7}$) and $n_{\mathrm{bg}} = 1.97 \times 10^6$ for the across view ($fpr_{\min} \approx 5.1 \times 10^{-7}$). Best TPR in each column is bolded. Reproducibility details are provided in Supplementary Material.

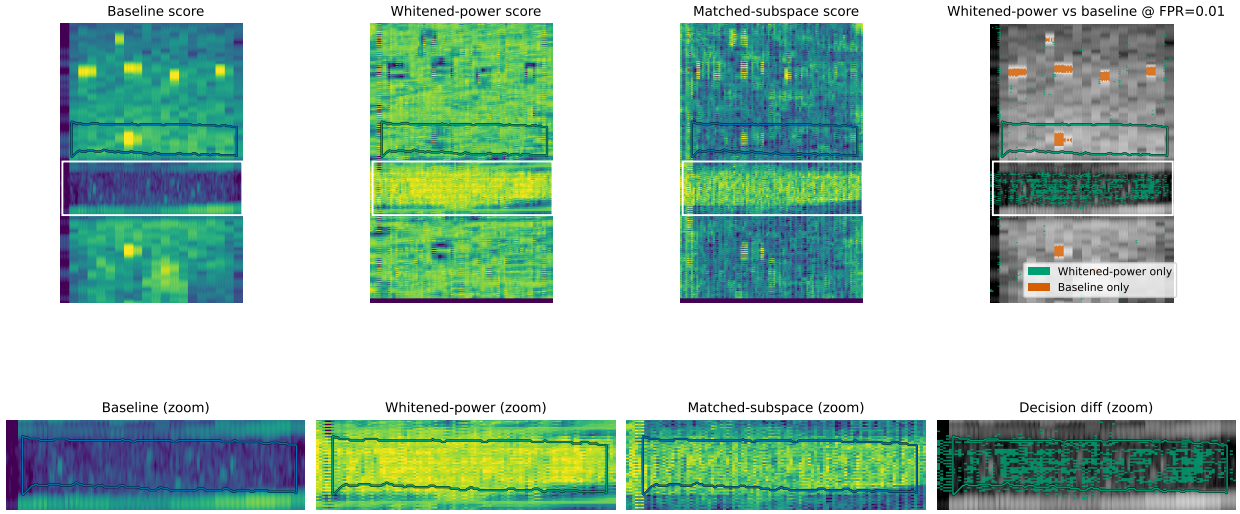

Figure 6: Leading structural-fidelity visual on a structurally labeled Gammex flow phantom acquisition from the open Doppler Twinkling Artifact database (labels derived from B-mode-only tube segmentation). Top row: baseline power Doppler image S_{base} (SVD band-pass clutter suppression followed by power Doppler), the whitened-power map, the matched-subspace map, and a matched-FPR decision-difference panel at a fixed background false-positive rate (10^{-2}; chosen here for visualization), showing pixels detected only by whitened power versus only by the baseline. Bottom row: zoomed region of interest. The tube contour is overlaid for reference. The qualitative ordering matches Table 5: whitened power is the strongest phantom score in this short-ensemble setting, while the matched-subspace panel remains included as the detector-core reference.

Additional Gammex sanity checks are provided in the Supplementary Material.

Within-ensemble motion ladders. A labeled within-ensemble motion ladder on the along-view Gammex cine is directionally consistent with the main phantom result and is useful as a robustness check, but the central phantom claim in the main text is the same-residual structural ROC comparison above.

5.4 ULM real-IQ audits

This section provides bounded real-IQ structural corroboration of the detector story, not the benchmark that determines the default variant.

Dataset and loader. We use a real kHz IQ dataset released for ultrasound localization microscopy benchmarking [38, 39] and distributed as HDF5 blocks with accompanying acquisition metadata. Each block contains a dataset `iq` of shape $(2, H, W, T) = (2, 256, 180, 800)$ float32 representing I/Q over $T = 800$ slow-time frames. We load blocks on demand and return complex IQ as a cube of shape (T, H, W) (reproduction details in Supplementary Material).

One-time label-free baseline selection. For ULM we select a single MC–SVD energy-fraction cutoff by a one-time label-free sweep on blocks 1–3 over frames 0:128, and then freeze it for all

subsequent ULM motion experiments. The sweep uses baseline-only runs and relies on label-free integrity and stability criteria (no task labels); it yields a frozen energy fraction $e = 0.975$ (full sweep details in Supplementary Material). The later PALA-backed structural evaluation and block-split threshold-transfer analysis use the same open dataset and therefore overlap in source blocks, but this one-time sweep used no localization-derived structural labels and fixed only the baseline MC–SVD energy fraction before those analyses were run.

5.4.1 Localization-derived structural evaluation

ULM-CoreShell. This section is a structural sanity check, not a functional claim: it asks whether detector scores concentrate more on known vessels than on the immediately adjacent parenchyma. To obtain a more anatomical structural readout on the same open real-IQ data, we use the published PALA rat-brain ULM example as an external vascular reference derived from microbubble localization microscopy rather than from the Doppler scores being evaluated [38,39]. We first reproduce the corresponding short-horizon PALA PowDop rendering from the Zenodo IQ blocks using the published odd-block selection and SVD cutoff, register that published PowDop crop back to the Zenodo IQ field of view, and then downsample the published super-resolution MatOut vascular rendering onto the short-window evaluation lattice. From that registered localization map we derive a conservative vessel-core versus perivascular-shell label pair: the vessel mask is a peak-based core mask, and the background mask is the surrounding shell, so the endpoint asks whether a short-window detector concentrates score on the vascular core rather than the immediately adjacent parenchyma. That one-time PALA-to-IQ registration is part of the label construction only. For the short-window detector itself we keep frame-to-frame registration disabled in the frozen audit, because re-enabling it changes the structural ROC only below the fourth decimal place while adding substantial latency.

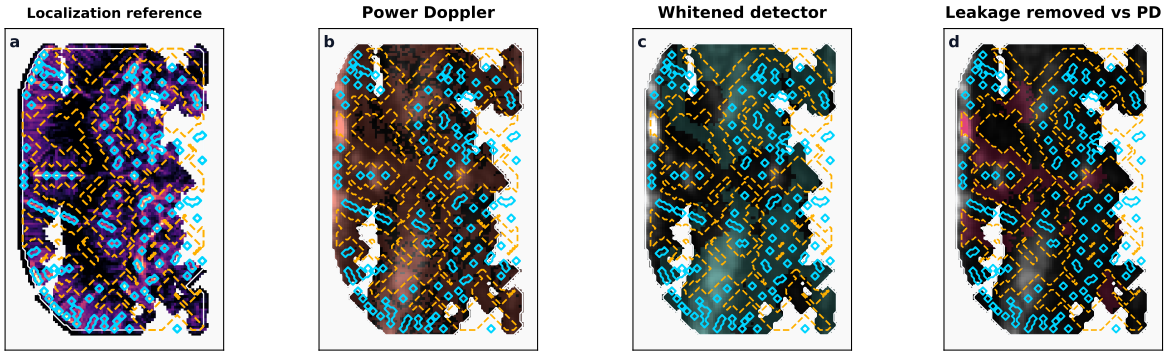

Figure 7: Localization-derived structural-label result on ULM Zenodo 7883227 using the published PALA rat-brain localization rendering as an external vascular reference. Left: the PALA-derived vessel-core and perivascular-shell masks after registration back to the IQ grid. Middle and right: the same audited 64-frame MC–SVD residual window (block 001, frames 128:192), thresholded at matched 70% vessel-core recall for baseline power Doppler and the prespecified no-registration whitened matched-subspace variant. Table 6 reports the formal same clutter-filtered-residual comparison of detection statistics.

Score	AUC (H1 vs H0)	TPR@10^{-2}	TPR@10^{-3}	shell FPR @70% core recall
Baseline (power Doppler)	0.504 [0.498,0.510]	0.010 [0.009,0.011]	7.9e-5 [0,2.4e-4]	0.698 [0.689,0.706]
Baseline (Kasai lag-1 magnitude)	0.503 [0.496,0.509]	0.009 [0.008,0.010]	5.9e-5 [0,1.8e-4]	0.701 [0.693,0.709]
Fixed matched-subspace statistic	0.514 [0.508,0.521]	0.014 [0.013,0.016]	2.6e-4 [1.2e-4,4.3e-4]	0.703 [0.694,0.711]
Adaptive statistic	0.514 [0.507,0.520]	0.014 [0.013,0.016]	2.6e-4 [1.2e-4,4.3e-4]	0.702 [0.693,0.711]
Fully whitened variant	**0.530 [0.524,0.537]**	**0.024 [0.021,0.027]**	**0.004 [0.003,0.006]**	**0.676 [0.665,0.687]**

Table 6: Same-residual ULM structural audit on real in vivo rat-brain IQ. All rows use the same frozen 64-frame MC–SVD residual cube and the same localization-derived vessel-core versus shell masks; only the downstream scoring rule changes. Entries report window-level means with 95% bootstrap intervals over the 70 audited windows. Best values in each metric column are bolded. The fully whitened variant exceeds baseline power Doppler on all 10 audited blocks (exact paired sign-flip test on block-mean AUC, p = 0.002).

On this endpoint, Table 6 reports PD, Kasai, the fixed statistic, the adaptive statistic, and the fully whitened variant on the same residual windows and the same localization-derived vessel-core versus shell masks. The fixed and adaptive statistics yield only modest pooled-ROC gains over PD and Kasai, and adaptive is nearly indistinguishable from fixed under this profile. The fully whitened variant is the only statistic that improves pooled AUC, strict-tail sensitivity, and shell FPR at matched 70% vessel-core recall: AUC 0.530 [0.524, 0.537], TPR@10^{-3} 0.004 [0.003, 0.006], and shell FPR 0.676 [0.665, 0.687], compared with AUC 0.514/0.514, TPR@10^{-3} 2.6e-4/2.6e-4, and shell FPR 0.703/0.702 for the fixed/adaptive statistics and AUC 0.504/0.503, TPR@10^{-3} 7.9e-5/5.9e-5, and shell FPR 0.698/0.701 for PD/Kasai.

The pooled advantage is not driven by one favorable block. Relative to the fixed statistic, the fully whitened variant exceeds AUC on 47/70 audited windows and all 10/10 blocks, lowers shell FPR at matched 70% core recall on 47/70 windows and 9/10 blocks, and yields nonzero TPR@10^{-3} in 45/70 windows versus 10/70 for the fixed and adaptive statistics. The same pattern holds against the adaptive statistic: 46/70 AUC windows, 47/70 shell-FPR windows, and 10/10 blocks. Against baseline power Doppler, the fully whitened variant also wins AUC on all 10 audited blocks on this same-acquisition ULM structural evaluation, and an exact paired sign-flip test on the block-mean AUC difference gives $p = 0.00195$. Figure 7 gives the visual counterpart to this result by showing a representative same-window comparison between baseline power Doppler and the fully whitened variant. On an RTX 4080 SUPER, the same 64-frame no-registration configuration runs at 41.3 ms steady-state total for a 64 ms acquisition budget under the reported replay conditions. The labels come from a published localization-derived vascular rendering on the same acquisition rather than from an independent modality or task-based ground truth, so this remains a structural rather than functional result.

To reduce overlap between localization-derived label construction and scored windows, we also ran a stricter odd-block-reference / even-block-evaluation variant under the same frozen 64-frame profile. The family ordering remained directionally consistent: the fully whitened variant still led on pooled AUC and strict-tail sensitivity, though with smaller margins than in the main same-acquisition audit. This stricter split reduces, but does not eliminate, the same-acquisition overlap limitation. Additional falsification controls point in the same direction: support-matched randomization of the vessel-core and shell masks collapses the structural audit toward chance, while a large cyclic translation destroys the fully whitened ordering rather than reproducing it.

A fixed-threshold audit on the same structural endpoint shows that PD and Kasai collapse to zero held-out vessel-core recall, whereas the fully whitened variant retains a small but nonzero

held-out core TPR under a transferred shell-calibrated threshold. That result remains modest, but it shows the structural claim does not depend entirely on per-window threshold readjustment.

5.4.2 Motion robustness audit

Motion audit. A motion-injection audit on ULM-CoreShell and Shin-ProxyMotion provides secondary robustness evidence under controlled residual-motion stress. In those audits, matched-subspace score maps are more stable than the baseline maps, but these results are corroboration rather than the structural efficacy claim that anchors the main real-IQ section.

Shin-ProxyMotion. To avoid making the motion-robustness story depend on a single open dataset, we also summarize the strongest existing Shin rat-brain motion-stress result from the supplement [36,37]. Under a prespecified low-frequency operating setting and a controlled residual-motion stress test, the matched-subspace map is more stable than the MC–SVD baseline map across all three audited windows, while the separate fixed-threshold proxy-transfer check remains negative.

The real-data results span four roles: Gammex-Struct provides structurally labeled phantom evidence, ULM-CoreShell adds a structural check using a localization-derived vascular reference plus block-split threshold transfer, ULM-Motion provides open real-IQ robustness evidence, and Shin-ProxyMotion provides a second open real-IQ corroboration under controlled residual-motion stress plus a negative cross-clip proxy-transfer check. Additional real-IQ audits, including same-cube score checks and extended motion variants, are summarized in the Supplementary Material.

5.5 Computational cost and replay feasibility

We also report representative runtimes for the final detector family evaluated in this manuscript rather than for a whitening-heavy research configuration. In this evaluation, the labeled-brain stress tests use the fixed matched-subspace default with conditional execution, the Shin real-IQ evaluation uses the adaptive variant, and Gammex uses the short-ensemble raw-baseline plus fixed matched-subspace statistic. Conditional execution changes the decision rule by deciding which tiles receive the adaptive detector, so it is reported as part of the detector configuration rather than as a free acceleration of the same method. Unless otherwise stated, conditional-execution reporting uses the same leakage-controlled, disjoint-window proxy discipline as the detection results.

Latency is measured by replaying multiple windows or frames in a single GPU process and reporting *steady-state* means over windows/frames 2..N. The first replay window/frame is excluded because it includes one-time startup overheads. We time the baseline and detector stages separately, excluding disk I/O, and "Total" reports baseline plus detector wall time. "Budget" denotes the acquired slow-time duration of the corresponding ensemble: $64/1500 \approx 42.7$ ms for the brain windows, $128/1000 = 128$ ms for Shin, and $17/2500 = 6.8$ ms for the Gammex cines. The reported real-time factor (RTF) is therefore Total/Budget.

In the final reported configurations, all three labeled-brain stress tests run below the full acquisition-PRF budget (RTF $\approx$ 0.86–0.88), the Shin real-IQ evaluation remains below budget (RTF ≈ 0.84), and both Gammex views run below the 17-shot cine-frame budget (RTF ≈ 0.88 along-view and ≈ 0.64 across-view).

A leakage ablation confirms that conditional execution can inflate apparent performance under same-window proxy gating, while leakage-controlled gating yields a small TPR decrease (order

10^{-3}) at the target FPRs. The full open-skull leakage table with bootstrap intervals is provided in the Supplementary Material.

6 Discussion

What the detector family adds

Post-beamformed fUS benefits from a better detection statistic, not only from a better clutter filter. The mechanism is setting-dependent: matched-subspace geometry drives the held-out SIMUS and labeled-brain gains, whereas local slow-time whitening becomes more important in short-ensemble or structure-concentration settings. Gammex-Struct is the clearest counterexample to a universal matched-subspace claim, because on the same phantom residual whitened power is stronger than the matched-subspace ratio. The result is therefore a detector family with a clear fixed default and narrower whitened variants, not one statistic that wins everywhere.

For a practitioner, the immediate implication is simple: the same SVD residual already used in an existing pipeline can yield a more reliable Doppler map by changing only the final detection statistic. On the held-out same-residual benchmark, the fixed statistic lowers intra-operative nuisance FPR from 0.998 for Kasai to 0.004 at matched 50% target-flow recall, while the separate best-chain benchmark yields zero observed nuisance detections on all four held-out seed-setting pairs. The strict-tail operating points are an evaluation lens, not a proposed clinical thresholding workflow; threshold transfer on raw *in vivo* IQ remains unresolved.

Real-data scope and clinical interpretation

Most public *in vivo* fUS releases do not preserve the raw high-PRF complex IQ and metadata needed for independent detector evaluation after beamforming. For that reason, the strongest efficacy evidence in this paper still comes from labeled simulations and structurally labeled phantom data, while the open real-IQ sections are narrower robustness and structural audits. The ULM-CoreShell result is directionally positive but modest, and only the fully whitened variant shows a clear advantage there. A stricter disjoint-source odd/even block audit remained directionally consistent but weaker, and a support-matched permutation control collapsed the structural audit toward chance. Together these checks reduce the simplest same-acquisition leakage objections, but they do not turn the result into independent multimodal or cross-session validation. The real-data contribution is therefore improved vascular and artifact discrimination at the detector stage, not biological inference or real-data functional task validation.

The held-out synthetic benchmark determines the default detector choice much more cleanly than the current real-IQ audits. What is still missing is a standardized open benchmark with repeated sessions and independent anatomical or task anchors for post-beamformed detector evaluation [8, 23–26].

Recommended detector variant by setting

Figure 8 summarizes the practical deployment rule supported by the current evidence. The fixed statistic remains the default, while the fully whitened variant is reserved for repeated guard-dominant windows with nonzero promotion and sufficient latency margin; the current adaptive telemetry is useful for label-free monitoring, but not yet a validated universal selector.

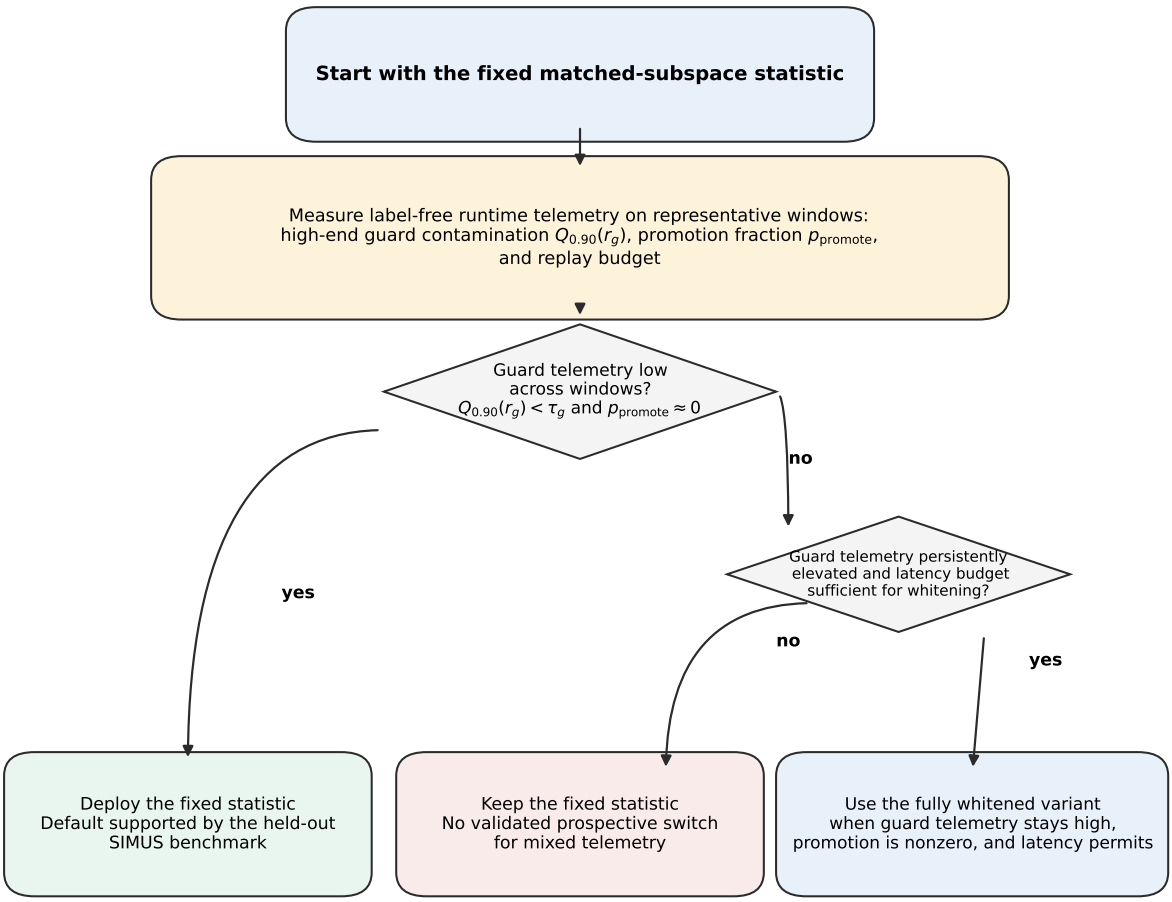

Figure 8: Conservative label-free deployment rule for choosing between the fixed and fully whitened detector variants. Here r_g measures guard-band contamination in a tile, $Q_{0.90}(r_g)$ is a high-end summary across a representative window, and p_{promote} is the fraction of tiles that would switch onto the whitened branch. Use the fixed detector by default and reserve the fully whitened variant for persistently clutter-dominant windows with sufficient latency margin. Mixed telemetry is not enough to justify an automatic switch.

Collaboration opportunities

The clearest next test is prospective validation on real task-driven fUS data. A collaborator would provide raw complex IQ or beamformed slow-time data, PRF and acquisition metadata, stimulus or task timing, and, when available, an independent structural anchor such as a localization-derived vascular map or structural co-registration. We would run the detector family on the collaborator's existing clutter-filtered residual pipeline, produce same-residual comparisons against PD/Kasai-style scores, and evaluate task-activation or structural readouts with the collaborator's current downstream analysis workflow rather than replacing it. The intended output is a joint validation study, positive or negative, together with a reusable evaluation harness the collaborator can con-

tinue using on future datasets. Collaborators would also receive the full detector implementation configured for their acquisition parameters, together with the evaluation harness, for use on future datasets. The contribution from this side is the tested detector implementation, the same-residual evaluation protocol, and the computational work needed to run the comparisons and package the results. The author welcomes collaboration inquiries from labs with raw fUS IQ data and task paradigms; correspondence to arthur@skymesasystems.com.

7 Conclusion and future work

We presented a localized matched-subspace detector framework for post-beamformed fUS and ultrafast Doppler data, centered on a band-limited fixed statistic and extended by an adaptive whitening switch plus a fully whitened variant. Across the evaluated labeled simulations, SIMUS-Struct, and one structurally labeled phantom setting, this detector family often improved low-FPR nuisance control relative to PD/Kasai-style downstream statistics, but the preferred downstream statistic depended on setting. Across those tiers, the fixed band-limited statistic remains the structural default, while the adaptive and fully whitened variants help in selected clutter-heavy settings rather than serving as universal replacements. The phantom result should be read accordingly: it supports localized covariance-aware downstream scoring, but not a universal matched-subspace mechanism.

On one open real-IQ rat-brain dataset, the same-acquisition ULM structural audit shows only modest gains for the fixed and adaptive statistics, while the fully whitened variant is the only statistic that clearly improves pooled AUC, strict-tail sensitivity, and shell suppression on the vessel-core versus perivascular-shell endpoint. A stricter disjoint-source odd/even block variant preserved that ordering with attenuated margins, so the real-IQ structural signal is not wholly dependent on block overlap inside the same acquisition. Fixed-threshold transfer remains mixed: SIMUS-Struct and the ULM structural audit retain non-retrospective signal, while the Shin proxy transfer stays negative. This is a detector-stage contribution for post-beamformed fUS pipelines, not completed real-data functional validation. The main next external test is prospective labeled *in vivo* validation on raw high-PRF IQ with independent anatomical or task anchors.

More broadly, freezing the clutter-suppressed cube and comparing only the downstream detection statistic provides a general protocol for benchmarking post-beamformed fUS detection methods. On existing open resources such as the PALA-backed ULM Zenodo archive and the Doppler Twinkling Artifact database, that same-residual layer can extend the utility of the released data by turning them into reusable post-beamformed benchmarks [38, 39, 43].

Data and code availability

All datasets used in this work are either publicly available (as cited) or derived deterministically from those sources. All code used to generate the reported tables and figures is versioned; the exact commit hash and release information are recorded in the reproducibility manifest (Supplementary Material). A minimal CUDA-enabled software environment and container recipe are provided in the repository. End-to-end regeneration commands for the reported figures and tables are collected in the repository and summarized in the reproducibility appendix.

Code repository. The reference implementation and reproduction scripts are available at `https://github.com/ArthurShune/fus-detectors`. The repository is released under the MIT License,

and the public codebase is the reference implementation used for the reported analyses. The canonical commit hash and release information are recorded in the reproducibility manifest (Supplementary Material).

References

[1] Imbault, M., et al. Intraoperative functional ultrasound imaging of human brain activity. *Scientific Reports* 7, 7304 (2017).

[2] Macé, E., Montaldo, G., Pernot, M., Cohen, I., and Tanter, M. Functional ultrasound imaging of the brain. *NeuroImage* 54(4), 2720–2731 (2011).

[3] Urban, A., et al. In vivo whole brain microvascular imaging in mice using ultrafast ultrasound. *EBioMedicine* 76, 103861 (2022).

[4] Alexandrov, A. V., et al. Velocity measurements in the middle cerebral arteries of healthy adults by transcranial Doppler. *Journal of Neuroimaging* 1, 166–172 (1991).

[5] Alexandrov, A. V., et al. Optimal values of flow velocity on transcranial Doppler in patients with intracranial arterial stenosis. *Stroke* 33, 1757–1761 (2002).

[6] Macé, E., et al. Functional ultrasound imaging of the brain: theory and basic principles. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 60(3), 492–506 (2013).

[7] Rabut, C., et al. Functional ultrasound imaging of human brain activity through the skull. *Science Translational Medicine* 12(572), eaav4392 (2020).

[8] Rabut, C., Norman, S. L., Griggs, W. S., Russin, J. J., Jann, K., et al. Functional ultrasound imaging of human brain activity through an acoustically transparent cranial window. *Science Translational Medicine* 16, eadj3143 (2024).

[9] Deffieux, T., Demené, C., and Tanter, M. Functional ultrasound imaging: A new imaging modality for neuroscience. *Neuroscience* 474, 110–121 (2021).

[10] Demeulenaere, O., Bertolo, A., Pezet, S., Osmanski, B.-F., Tanter, M., and Deffieux, T. In vivo whole brain microvascular imaging in mice using transcranial 3D ultrasound localization microscopy. *EBioMedicine* 79, 103995 (2022).

[11] Demené, C., et al. Spatiotemporal clutter filtering of ultrafast ultrasound data highly increases Doppler and fUltrasound sensitivity. *IEEE Transactions on Medical Imaging* (2015). DOI: `https://doi.org/10.1109/TMI.2015.2428634`.

[12] Baranger, J., et al. Adaptive spatiotemporal SVD clutter filtering for ultrafast Doppler imaging using similarity of spatial singular vectors. *IEEE Transactions on Medical Imaging* (2018). DOI: `https://doi.org/10.1109/TMI.2018.2789499`.

[13] Treeby, B. E., and Cox, B. T. k-Wave: MATLAB toolbox for the simulation and reconstruction of photoacoustic wave fields. *Journal of Biomedical Optics* 15(2), 021314 (2010). DOI: `https://doi.org/10.1117/1.3360308`.

[14] Yu, A. C. H., and Løvstakken, L. Eigen-based clutter filter design for ultrasound color flow imaging: a review. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2010). DOI: https://doi.org/10.1109/TUFFC.2010.1521.

[15] Song, P., et al. Ultrasound small vessel imaging with block-wise adaptive local clutter filtering. *IEEE Transactions on Medical Imaging* (2017). DOI: https://doi.org/10.1109/TMI.2016.2605819.

[16] Kasai, C., Namekawa, K., Koyano, A., and Omoto, R. Real-time two-dimensional blood flow imaging using an autocorrelation technique. *IEEE Transactions on Sonics and Ultrasonics* 32(3), 458–464 (1985). DOI: https://doi.org/10.1109/T-SU.1985.31615.

[17] Chee, A. J. Y., Yiu, B. Y. S., and Yu, A. C. H. A GPU-parallelized eigen-based clutter filter framework for ultrasound color flow imaging. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2017). DOI: https://doi.org/10.1109/TUFFC.2016.2606598.

[18] Lok, U.-W., Tang, J., Kilic, K., Szabo, T. L., and Boas, D. A. Real time SVD-based clutter filtering using randomized SVD and spatial downsampling for microvessel imaging. *Ultrasonics* 107, 106163 (2020). DOI: https://doi.org/10.1016/j.ultras.2020.106163.

[19] Demene, C., et al. Improved color Doppler for cerebral blood flow axial velocity estimation (iCD-fUS). *IEEE Transactions on Medical Imaging* 40(5), 1498–1510 (2021).

[20] Tang, J., Kilic, K., Szabo, T. L., and Boas, D. A. Improved color Doppler for cerebral blood flow axial velocity imaging. *IEEE Transactions on Medical Imaging* 40(2), 758–764 (2021).

[21] Demené, C., et al. Functional ultrasound imaging of brain activity in human newborns. *Science Translational Medicine* 9(411), eaah6756 (2017). DOI: https://doi.org/10.1126/scitranslmed.aah6756.

[22] Baranger, J., et al. Bedside functional monitoring of the dynamic brain connectivity in human neonates. *Nature Communications* 12, 1080 (2021). DOI: https://doi.org/10.1038/s41467-021-21387-x.

[23] Soloukey, S., et al. Mobile human brain imaging using functional ultrasound. *Science Advances* 11(25), eadu9133 (2025). DOI: https://doi.org/10.1126/sciadv.adu9133.

[24] Lambert, T., Brunner, C., Urban, A., et al. Functional ultrasound imaging and neuronal activity: How accurate is the spatiotemporal match? *Imaging Neuroscience* 3, IMAG.a.34 (2025). DOI: https://doi.org/10.1162/IMAG.a.34.

[25] Edelman, B. J., et al. The COMBO window: A chronic cranial implant for multiscale circuit interrogation in mice. *PLOS Biology* 22(6), e3002664 (2024). DOI: https://doi.org/10.1371/journal.pbio.3002664.

[26] Chen, Z., et al. Functional ultrasound imaging of auditory responses in comatose patients. *Research* 8, 0709 (2025). DOI: https://doi.org/10.34133/research.0709.

[27] Nunez-Elizalde, A. O., et al. Neural correlates of blood flow measured by ultrasound. *Neuron* 110(10), 1631–1640.e4 (2022). DOI: https://doi.org/10.1016/j.neuron.2022.02.012.

[28] Aaslid, R., Markwalder, T.-M., and Nornes, H. Noninvasive transcranial Doppler ultrasound recording of flow velocity in basal cerebral arteries. *Journal of Neurosurgery* 57(6), 769–774 (1982).

[29] Gao, S., Lam, W. W.-K., Chan, Y.-H., Liu, J., Wong, K.-P., Leung, T. W.-S., et al. Optimal values of flow velocity on transcranial Doppler in grading middle cerebral artery stenosis in comparison with magnetic resonance angiography. *Journal of Neuroimaging* 12(3), 213–218 (2002).

[30] Estrada, H., Gottschalk, S., Reiss, M., Neuschmelting, V., Razansky, D., et al. Observation of guided acoustic waves in a human skull. *Ultrasound in Medicine & Biology* 44(7), 1438–1447 (2018).

[31] Candès, E. J., Li, X., Ma, Y., and Wright, J. Robust principal component analysis? *Journal of the ACM* 58(3), 11 (2011).

[32] Solomon, O., et al. Deep unfolded robust PCA with application to clutter suppression in ultrasound. *IEEE Transactions on Medical Imaging* 39(4), 1051–1063 (2020). DOI: `https://doi.org/10.1109/TMI.2019.2941271`.

[33] Lin, Z., Chen, M., and Ma, Y. The augmented Lagrange multiplier method for exact recovery of corrupted low-rank matrices. *arXiv preprint* arXiv:1009.5055 (2010).

[34] Halko, N., Martinsson, P. G., and Tropp, J. A. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review* 53(2), 217–288 (2011).

[35] Soloukey, S., Vincent, A. J. P. E., Smits, M., Slump, C. H., Dirven, C. M. F., et al. Functional ultrasound (fUS) during awake brain surgery: the clinical potential for intra-operative functional and vascular brain mapping. *Frontiers in Neuroscience* 13, 1384 (2019). DOI: `https://doi.org/10.3389/fnins.2019.01384`.

[36] Shin, Y., et al. Context-Aware Deep Learning Enables High-Efficacy Localization of High Concentration Microbubbles for Super-Resolution Ultrasound Localization Microscopy. Zenodo dataset, DOI 10.5281/zenodo.10711806 (2024).

[37] Shin, Y., et al. Context-Aware Deep Learning Enables High-Efficacy Localization of High Concentration Microbubbles for Super-Resolution Ultrasound Localization Microscopy. *Nature Communications* 15, 3299 (2024). DOI: `https://doi.org/10.1038/s41467-024-47154-2`.

[38] Chavignon, A., Heiles, B., Hingot, V., Lopez, P., Teston, E., and Couture, O. In vivo rat brain for Ultrasound Localization Microscopy: raw and beamformed data. Zenodo dataset, DOI 10.5281/zenodo.7883227 (2023).

[39] Heiles, B., Chavignon, A., Hingot, V., Lopez, P., Teston, E., and Couture, O. Performance benchmarking of microbubble-localization algorithms for ultrasound localization microscopy. *Nature Biomedical Engineering* (2021). DOI: `https://doi.org/10.1038/s41551-021-00824-8`.

[40] Errico, C., et al. Ultrafast ultrasound localization microscopy for deep super-resolution vascular imaging. *Nature* 527(7579), 499–502 (2015). DOI: https://doi.org/10.1038/nature16066.

[41] Hingot, V., et al. Subwavelength motion-correction for ultrafast ultrasound localization microscopy. *Ultrasonics* 77, 17–21 (2017). DOI: https://doi.org/10.1016/j.ultras.2017.01.008.

[42] Insana, M. F., Dai, B., Babaei, S., and Abbey, C. K. Combining spatial registration with clutter filtering for power-Doppler imaging in peripheral perfusion applications. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 69(12), 3243–3254 (2022). DOI: https://doi.org/10.1109/TUFFC.2022.3211469.

[43] Leonov, D. V., Reshetnikov, R. V., Kulberg, N. S., Nasibullina, A. A., and Gromov, A. I. Doppler Twinkling Artifact Observations: an Open-Access Database of Raw Ultrasonic Signals. *Digital Diagnostics* (2021). DOI: https://doi.org/10.17816/DD76511.

[44] Ozgun, K., and Byram, B. C. Multidimensional clutter filtering of aperture domain data for improved blood flow sensitivity. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 68(8), 2645–2656 (2021). DOI: https://doi.org/10.1109/TUFFC.2021.3073292.

[45] Huang, Z., et al. Adaptive higher-order singular value decomposition clutter filter for ultrafast Doppler imaging of coronary flow under non-negligible tissue motion. *Ultrasonics* (2024). DOI: https://doi.org/10.1016/j.ultras.2024.107307.

[46] Garcia, D. SIMUS: an open-source simulator for medical ultrasound imaging. Part I: theory and examples. *Computer Methods and Programs in Biomedicine* 218, 106726 (2022). DOI: https://doi.org/10.1016/j.cmpb.2022.106726.

[47] Cigier, A., Varray, F., and Garcia, D. SIMUS: an open-source simulator for medical ultrasound imaging. Part II: comparison with four simulators. *Computer Methods and Programs in Biomedicine* 220, 106774 (2022). DOI: https://doi.org/10.1016/j.cmpb.2022.106774.

[48] Urban, A., et al. Real-time imaging of brain activity in freely moving rodents using functional ultrasound. *Nature Methods* 12(9), 873–878 (2015). DOI: https://doi.org/10.1038/nmeth.3482.

[49] Demené, C., et al. 4D microvascular imaging based on ultrafast Doppler tomography. *NeuroImage* 127, 472–483 (2016). DOI: https://doi.org/10.1016/j.neuroimage.2015.11.014.

[50] Slepian, D. Prolate spheroidal wave functions, Fourier analysis, and uncertainty—V: The discrete case. *Bell System Technical Journal* 57(5), 1371–1430 (1978). DOI: https://doi.org/10.1002/j.1538-7305.1978.tb02104.x.

[51] Thomson, D. J. Spectrum estimation and harmonic analysis. *Proceedings of the IEEE* 70(9), 1055–1096 (1982). DOI: https://doi.org/10.1109/PROC.1982.12433.

[52] Percival, D. B., and Walden, A. T. *Spectral Analysis for Physical Applications: Multitaper and Conventional Univariate Techniques.* Cambridge University Press (1993).

[53] Guerci, J. R. Artech House monograph (2003).

[54] Wicks, M. C., Rangaswamy, M., Adve, R., and Hale, T. B. *IEEE Signal Processing Magazine* 23(1), 51–65 (2006). DOI: `https://doi.org/10.1109/MSP.2006.1593337`.

[55] Melvin, W. L., and Guerci, J. R. *IEEE Transactions on Aerospace and Electronic Systems* 42(3), 983–996 (2006). DOI: `https://doi.org/10.1109/TAES.2006.248215`.

[56] Gurram, P. R., and Goodman, N. A. Spectral-domain covariance estimation with a priori knowledge. *IEEE Transactions on Aerospace and Electronic Systems* 42, 1010–1020 (2006). DOI: `https://doi.org/10.1109/TAES.2006.248209`.

[57] Van Veen, B. D., and Buckley, K. M. Beamforming: a versatile approach to spatial filtering. *IEEE ASSP Magazine* 5(2), 4–24 (1988). DOI: `https://doi.org/10.1109/53.665`.

[58] Kuglin, C. D., and Hines, D. C. The phase correlation image alignment method. In *Proceedings of the IEEE International Conference on Cybernetics and Society* (1975).

[59] Stone, H. S., Orchard, M. T., Chang, E.-C., and Martucci, S. A. A fast direct Fourier-based algorithm for subpixel registration of images. *IEEE Transactions on Geoscience and Remote Sensing* 39(10), 2235–2243 (2001). DOI: `https://doi.org/10.1109/36.957286`.

[60] Foroosh, H., Zerubia, J. B., and Berthod, M. Extension of phase correlation to subpixel registration. *IEEE Transactions on Image Processing* 11(3), 188–200 (2002). DOI: `https://doi.org/10.1109/83.988953`.

[61] Capon, J. High-resolution frequency-wavenumber spectrum analysis. *Proceedings of the IEEE* 57(8), 1408–1418 (1969).

[62] Frost, O. L. An algorithm for linearly constrained adaptive array processing. *Proceedings of the IEEE* 60(8), 926–935 (1972).

[63] Kay, S. M. *Fundamentals of Statistical Signal Processing, Volume II: Detection Theory.* Prentice Hall (1998).

[64] Scharf, L. L., and Friedlander, B. Matched subspace detectors. *IEEE Transactions on Signal Processing* 42(8), 2146–2157 (1994).

[65] Ward, J. MIT Lincoln Laboratory, Tech. Rep. 1015 (1994).